

Constraint-threshold conjunction: Real Stability

Research Architecture Runtime
operator/constraint-threshold-conjunction

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Abstract

`constraint-threshold-conjunction` preserves a Boolean combination of threshold constraints under buffered scalar noise. Lean `LEAN_FULL`.

This theorem is: Reduction

1 Problem

Conjunction or disjunction of threshold predicates on coordinates/scores.

2 Stability notion

Each literal is threshold-stable; Boolean combinations inherit preservation.

3 Definitions

Reduce to multi-threshold / threshold preservation on each cut.

4 Theorem

Theorem 1. *If every atomic threshold is buffer-stable for (x, x', ε) , the conjunction/disjunction value is preserved.*

5 Intuition

Boolean operations are deterministic functions of stable bits.

6 Examples

Example 2. Two cuts $x \geq 1$ and $x \geq 3$ with $x = 4$, $\varepsilon = 0.2$: both bits stable, so their conjunction stays true.

7 Proof

Each atomic predicate is preserved by scalar threshold preservation under the buffer hypotheses. Conjunction (resp. disjunction) is a fixed Boolean function of those bits, hence preserved (Theorem 1).

8 Formal statement

Lean module

```
Research.Operators.ConstraintThresholdConjunction.Preservation
```

Source file

```
lean/Research/Operators/ConstraintThresholdConjunction/Preservation.  
lean
```

Theorems

- `constraint\_threshold\_conjunction\_conjunction\_preservation`
- `constraint\_threshold\_conjunction\_conjunction\_sharpness`

Certificate

```
lean/certificates/constraint-threshold-conjunction/constraint-thresho  
ld-conjunction-conjunction-preservation
```

Status

```
LEAN_FULL on Mathlib  $\mathbb{R}$  (REAL_MATHLIB)
```

9 Proof dependencies

- Threshold / multi-threshold cores.
- Boolean congruence.

10 Consequences

Constraint-side companion to multi-threshold counts.